# ERDŐS PROBLEM 741 (LOWER-DENSITY VARIANT): POSITIVE LOWER DENSITY OF $A+A$ DOES NOT PASS TO A PARTITION

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## 1. The problem

For a set $S \subseteq \mathbb{N} = \{0, 1, 2, \dots\}$, write

$$\underline{d}(S) = \liminf_{n\to\infty} \frac{|S \cap [0,n)|}{n} \qquad \text{and} \qquad \overline{d}(S) = \limsup_{n\to\infty} \frac{|S \cap [0,n)|}{n}$$

for its lower and upper asymptotic densities, and write $A+A = \{a+b : a, b \in A\}$ for the sumset. In [1] (see [2]), Erdős asked whether every set $A \subseteq \mathbb{N}$ with $\overline{d}(A+A) > 0$ can be partitioned as $A = A_1 \sqcup A_2$ so that $\overline{d}(A_1+A_1) > 0$ and $\overline{d}(A_2+A_2) > 0$. The formal-conjectures benchmark [3] records the companion question for lower density: if $\underline{d}(A+A) > 0$, can one always partition $A = A_1 \sqcup A_2$ so that $\underline{d}(A_1+A_1) > 0$ and $\underline{d}(A_2+A_2) > 0$? The answer is no.

**Theorem.** *There exists a set $A \subseteq \mathbb{N}$ with $\underline{d}(A+A) \geq 3/256 > 0$ such that for every disjoint decomposition $A = A_1 \sqcup A_2$,*

$$\min\big(\underline{d}(A_1+A_1),\, \underline{d}(A_2+A_2)\big) = 0.$$

## 2. Idea of the construction

Why does the upper-density problem behave differently from the lower-density one? To have $\overline{d}(A_i+A_i) > 0$, each piece $A_i$ need only fill a constant fraction of *some* infinitely many prefixes $[0,N)$, so $A_1$ and $A_2$ can take turns across alternating scales. By contrast, $\underline{d}(A_1+A_1) > 0$ and $\underline{d}(A_2+A_2) > 0$ require *both* sumsets to have density bounded away from zero at *every* sufficiently large prefix $[0,N)$ simultaneously.

We exploit this by building $A$ as a union of disjoint stages $G_k \subseteq [M_k, M_{k+1})$ $(k = 0, 1, 2, \dots)$ with a parameter $m = m_k = 2k+4 \to \infty$. At stage $k$:

- A finite *hub* set $H_k$ of $2m^2$ elements is organized into $m$ blocks of $2m$ elements each. Any 2-coloring $A = A_1 \sqcup A_2$ must give one color (say $A_1$) at least $m$ elements in at least $q = m/2$ blocks. Such a selection of $q \times m$ hubs is called a *configuration* $\gamma$.
- There are finitely many $(R_k)$ configurations $\gamma_0, \dots, \gamma_{R_k-1}$, and we allocate each $\gamma_r$ its own geometric scale window $[N_{k,r}, N_{k,r+1})$ with $N_{k,r+1} = 4mN_{k,r}$.
- On $[N_{k,r}, N_{k,r+1})$, we place a sparse *spoke* set $W_{k,r}$ constrained by two independent conditions: every $w \in W_{k,r}$ lies in a short residue window $w \bmod P_k < D_{0,k}/2$ (occupying a $1/(4m)$ fraction of residues modulo $P_k$), and satisfies a base-$B_k$ linear congruence $\Lambda_{\gamma_r}(w) \equiv 0 \pmod m$ whose coefficients on digit positions are tuned to $\gamma_r$.
- Adding the $qm = m^2/2$ hubs of $\gamma_r$ to $W_{k,r}$ shifts both the coarse residue modulo $P_k$ (into $q$ distinct $D_{0,k}$-slots) and the linear form $\Lambda_{\gamma_r} \bmod m$ (through all $m$ residues $0, \dots, m-1$) without base-$B_k$ carries, expanding $W_{k,r}$ injectively to cover a bounded-below fraction of $[N_{k,r}, N_{k,r+1})$. Auxiliary intervals $I_k$ and grids $J_k$ bridge the transition between $M_k$ and $N_{k,0}$, keeping $|(A+A) \cap [0,n)| \geq \frac{3}{256}n$ for *every* $n \geq 16$.
- However, at the specific scale $N = N_{k,r+1}$ corresponding to a monochromatic configuration $\gamma_r \subseteq A_1$, the *other* piece $A_2$ contains *none* of the hubs of $\gamma_r$. Every remaining hub in $H_k \cap A_2$ has coefficient 0 in $\Lambda_{\gamma_r}$, so $(A_2+A_2) \cap [0,N)$ is trapped inside the union of

$[0, 2N_{k,r})$ (density $1/(2m)$), the small modulo-$P_k$ window $< D_{0,k}$ (density $1/(2m)$), and the kernel $\Lambda_{\gamma_r}(z) \equiv 0 \pmod m$ (density $1/m$). Thus

$$|(A_2 + A_2) \cap [0, N_{k,r+1})| \le \frac{2}{m_k} N_{k,r+1},$$

forcing $\min\big(\underline{d}(A_1 + A_1), \underline{d}(A_2 + A_2)\big) \le \lim_{k\to\infty} 2/m_k = 0$.

## 3. Base-$B$ linear-form gadget

For integers $B \ge 2$, $x \ge 0$, and $t \ge 0$, let $\mathrm{dig}_B(x, t) = \lfloor x/B^t \rfloor \bmod B$ be the $t$-th base-$B$ digit of $x$. Two elementary properties govern carry-free digit shifts:

(1) If $B^d \mid z$ and $t < d$, then $\mathrm{dig}_B(z + w, t) = \mathrm{dig}_B(w, t)$.
(2) If $\mathrm{dig}_B(w, t) < B - 1$, then adding $B^t$ to $w$ produces no carry into position $t + 1$, so $\mathrm{dig}_B(w + B^t, t') = \mathrm{dig}_B(w, t') + \mathbf{1}_{t'=t}$ for every $t' \ge 0$.

Fix a stage $k \ge 0$ and set

$$q = k + 2, \quad m = 2q = 2k + 4 \ge 4, \quad d_0 = 2m^2, \quad d = d_0 + 1, \quad B = 8m^3, \quad Q = B^d.$$

Index the $d_0 = m \cdot (2m)$ pairs $(j, u) \in \{0, \dots, m-1\} \times \{0, \dots, 2m-1\}$ by a bijection $\phi : \{0, \dots, m-1\} \times \{0, \dots, 2m-1\} \xrightarrow{\sim} \{0, \dots, d_0 - 1\}$.

A *configuration* $\gamma = (\iota, \sigma)$ consists of an injective map $\iota : \{0, \dots, q-1\} \hookrightarrow \{0, \dots, m-1\}$ together with, for each $i < q$, an injective map $\sigma_i : \{0, \dots, m-1\} \hookrightarrow \{0, \dots, 2m-1\}$. There are $R_k \ge 1$ such configurations; enumerate them as $\gamma_0, \dots, \gamma_{R_k - 1}$.

For a configuration $\gamma = (\iota, \sigma)$, assign each pair $(j, u)$ the weight

$$c_\gamma(j, u) = \begin{cases} v & \text{if } (j, u) = (\iota(i), \sigma_i(v)) \text{ for some } i < q \text{ and } v < m, \\ 0 & \text{otherwise} \end{cases}$$

(well-defined by injectivity of $\iota$ and $\sigma_i$), and define $\Lambda_\gamma : \mathbb{N} \to \mathbb{N}$ by

$$\Lambda_\gamma(x) = (x \bmod B) + \sum_{j<m} \sum_{u<2m} c_\gamma(j, u)\, \mathrm{dig}_B\big(\lfloor x/B \rfloor, \phi(j, u)\big).$$

Notice that $\Lambda_\gamma$ depends only on $x \bmod B$ and the first $d_0$ base-$B$ digits of $\lfloor x/B \rfloor$, i.e. on $x \bmod B^{d_0+1} = x \bmod Q$. Hence $\Lambda_\gamma(z + w) = \Lambda_\gamma(w)$ whenever $Q \mid z$.

**Lemma 1** (Density of carry-safe kernel elements)**.** *Let $S_\gamma$ be the set of integers $x \in [0, Q)$ such that $\mathrm{dig}_B(\lfloor x/B \rfloor, s) < B - 1$ for all $s < d_0$ and $\Lambda_\gamma(x) \equiv 0 \pmod m$. Then*

$$|S_\gamma| \ge \frac{3Q}{4m}.$$

*Proof.* There are $(B-1)^{d_0}$ choices of $p_1 \in [0, B^{d_0})$ whose digits in positions $0, \dots, d_0 - 1$ are all $< B - 1$. For each such $p_1$ and each $p_2 \in \{0, \dots, 8m^2 - 1\}$, there is a unique residue $r_0 \in \{0, \dots, m-1\}$ making $p_2 m + r_0 + \sum c_\gamma(j, u)\, \mathrm{dig}_B(p_1, \phi(j, u)) \equiv 0 \pmod m$. Since $p_2 m + r_0 < 8m^3 = B$, the numbers $x = p_1 B + p_2 m + r_0$ are pairwise distinct and lie in $S_\gamma$, so $|S_\gamma| \ge (B-1)^{d_0} \cdot 8m^2$. By Bernoulli's inequality $B^{d_0} \le (B-1)^{d_0} + d_0 B^{d_0 - 1}$ and $4d_0 = 8m^2 \le B$, we have $(B-1)^{d_0} \ge \frac{3}{4} B^{d_0}$, whence

$$|S_\gamma| \ge \frac{3}{4} B^{d_0} \cdot 8m^2 = \frac{3B^{d_0+1}}{4m} = \frac{3Q}{4m}. \qquad \square$$

## 4. Multi-scale counterexample and $\underline{d}(A + A) > 0$

Define $M_0 = 16$ and, given $M_k$ at stage $k$, set

$$L_{0,k} = 4M_k Q_k, \qquad D_{0,k} = 4m_k L_{0,k}, \qquad P_k = 2m_k D_{0,k}, \qquad N_{0,k} = 10 m_k P_k = 80 m_k^3 L_{0,k},$$
$$N_{k,r} = (4m_k)^r N_{0,k} \quad (0 \le r \le R_k), \qquad M_{k+1} = N_{k,R_k}.$$

Note the exact divisibility chain $B_k \mid Q_k \mid L_{0,k} \mid (D_{0,k}/2) \mid D_{0,k} \mid P_k \mid N_{0,k} \mid N_{k,r}$. At stage $k$, define four subsets of $[M_k, M_{k+1})$:

(1) the *interval* $I_k = [M_k, L_{0,k})$;

(2) the *bridge grid* $J_k = \{tL_{0,k} : 1 \le t < 80m_k^3\}$;
(3) the *hubs* $H_k = \{h_k(j,u) : j < m_k,\ u < 2m_k\}$, where

$$h_k(j,u) = (j+1)D_{0,k} + B_k^{\phi_k(j,u)+1} \in [D_{0,k}, P_k),$$

and write $H_k(\gamma) = \{h_k(\iota(i), \sigma_i(v)) : i < q_k,\ v < m_k\}$ for the $q_k m_k$ hubs of $\gamma = (\iota, \sigma)$;
(4) for each $r < R_k$, the *spokes*

$$W_{k,r} = \big\{w \in [N_{k,r}, N_{k,r+1}) : w \bmod P_k < D_{0,k}/2,\ m_k \mid \Lambda_{\gamma_r}(w), \\ \forall s < d_{0,k},\ \mathrm{dig}_{B_k}(\lfloor w/B_k \rfloor, s) < B_k - 1\big\}.$$

Put $G_k = I_k \cup J_k \cup H_k \cup \bigcup_{r<R_k} W_{k,r} \subseteq [M_k, M_{k+1})$ and $A = [0,16) \cup \bigcup_{k\ge 0} G_k$.

If $w$ has all digits $\mathrm{dig}_{B_k}(\lfloor w/B_k \rfloor, s) < B_k - 1$ $(s < d_{0,k})$, then since $Q_k \mid (j+1)D_{0,k}$ and $B_k^{\phi_k(j,u)+1} = B_k^{\phi_k(j,u)} \cdot B_k$ increments digit $\phi_k(j,u)$ of $\lfloor w/B_k \rfloor$ by 1 without carry,

$$\Lambda_\gamma\big(h_k(j,u) + w\big) = \Lambda_\gamma(w) + c_\gamma(j,u). \tag{1}$$

**Lemma 2** (Lower density of $A+A$)**.** *For every integer $n \ge 16$, $|(A+A) \cap [0,n)| \ge \frac{3}{256}n$. In particular, $\underline{d}(A+A) \ge 3/256$.*

*Proof.* We first establish $|(A+A) \cap [0,n)| \ge \frac{3}{64}n$ on the two main regimes of stage $k$:

- *Scale regime $4N_{k,r} \le n \le N_{k,r+1}$ $(r < R_k)$.* Write $\gamma_r = (\iota, \sigma)$, $p_0 = N_{k,r}/P_k \ge 40$, and $Q_n = \lfloor n/P_k \rfloor \ge 4p_0 \ge 160$. Consider all sums

$$z = h_k(\iota(i), \sigma_i(v)) + \big((p_0 + t)P_k + bQ_k + x\big)$$

with $i < q_k$, $v < m_k$, $0 \le t < Q_n - p_0 - 1$, $0 \le b < 8m_kM_k = D_{0,k}/(2Q_k)$, and $x \in S_{\gamma_r}$ (from Lemma 1). Because $Q_k \mid P_k$ and $bQ_k + x < D_{0,k}/2$, the second summand $w$ belongs to $W_{k,r}$ and $z < (p_0 + t + 2)P_k \le n$. Moreover, $(i, v, t, b, x)$ is uniquely recovered from $z$:
  (1) since $h_k + bQ_k + x < (m_k+1)D_{0,k} + D_{0,k}/2 < P_k$, $\lfloor z/P_k \rfloor = p_0 + t$ determines $t$;
  (2) since $B_k^{\phi_k+1} + bQ_k + x < Q_k + D_{0,k}/2 < D_{0,k}$, $\lfloor (z \bmod P_k)/D_{0,k} \rfloor = \iota(i) + 1$ determines $i$;
  (3) by (1), $\Lambda_{\gamma_r}(z) \bmod m_k = (0 + v) \bmod m_k = v$ determines $v$;
  (4) subtracting $h_k(\iota(i), \sigma_i(v)) + (p_0 + t)P_k$ leaves $bQ_k + x$ with $x < Q_k$, determining $b$ and $x$.

  By Lemma 1, the number of such sums is at least

$$q_k m_k (Q_n - p_0 - 1)(8m_kM_k)\frac{3Q_k}{4m_k} \ge \frac{Q_n + 1}{2} \cdot 3m_k^3 M_k Q_k = \frac{3}{64}(Q_n + 1)P_k > \frac{3}{64}n.$$

- *Bridge regime $2L_{0,k} \le n \le N_{0,k}$.* Put $Q_L = \lfloor n/L_{0,k} \rfloor \in [2, 80m_k^3]$. The sums $tL_{0,k} + x$ with $1 \le t < Q_L$ $(tL_{0,k} \in J_k)$ and $x \in [M_k, L_{0,k}) = I_k$ are pairwise distinct and $< n$, giving $(Q_L - 1)(L_{0,k} - M_k) \ge \frac{Q_L+1}{3} \cdot \frac{3}{4}L_{0,k} > \frac{1}{4}n > \frac{3}{64}n$ elements.

Now let $n \ge 16$ and choose $k \ge 0$ with $M_k \le n < M_{k+1}$. Note that $0 \in [0,16) \subseteq A$.

- If $4M_k \le n \le 2L_{0,k}$, the sums $0 + x$ with $x \in [M_k, \min(n, L_{0,k})) \subseteq I_k$ give $\min(n, L_{0,k}) - M_k \ge \frac{1}{8}n > \frac{3}{256}n$ elements.
- If $n \in [M_k, 4M_k)$ with $k = 0$, $0 + 0 = 0 < n < 64$ gives $1 > \frac{3}{256}n$. If $n \in [M_k, 4M_k)$ with $k \ge 1$ (where $M_k = N_{k-1,R_{k-1}}$) or $n \in [N_{k,r}, 4N_{k,r})$ $(r < R_k)$, the left endpoint $n' \in \{M_k, N_{k,r}\}$ already satisfies $|(A+A) \cap [0,n')| \ge \frac{3}{64}n'$ by the two regimes above, so $|(A+A) \cap [0,n)| \ge \frac{3}{64}n' > \frac{3}{256}n$. □

## 5. Partition pigeonhole and sumset bottleneck

**Lemma 3** (Monochromatic configuration)**.** *For every decomposition $A = A_1 \cup A_2$ and every $k \ge 0$, there exists $r < R_k$ such that $H_k(\gamma_r) \subseteq A_1$ or $H_k(\gamma_r) \subseteq A_2$.*

*Proof.* In each block $j < m_k$, the $2m_k$ hubs $\{h_k(j,u) : u < 2m_k\} \subseteq A_1 \cup A_2$ give at least $m_k$ hubs to $A_1$ or at least $m_k$ hubs to $A_2$. Since there are $m_k = 2q_k$ blocks, one of $A_1, A_2$ wins at least $q_k$ blocks, which contains $H_k(\gamma_r)$ for some configuration $\gamma_r$. □

**Lemma 4** (Sumset bottleneck)**.** *Suppose $A = A_1 \sqcup A_2$ and $H_k(\gamma_r) \subseteq A_o$, where $\{A_t, A_o\} = \{A_1, A_2\}$. Then at $N = N_{k,r+1}$,*

$$|(A_t + A_t) \cap [0, N)| \leq \frac{2N}{m_k}.$$

*Proof.* Inspect $A_t \cap [0, N)$. Since $G_{k'} \subseteq [M_{k'}, M_{k'+1})$ and $N \leq M_{k+1}$, no stage $k' > k$ meets $[0, N)$, nor does $W_{k,u}$ for $u > r$. Every element of $[0, 16) \cup \bigcup_{k'<k} G_{k'} \cup I_k$ is $< L_{0,k} < D_{0,k}/2 < P_k < N_{k,r}$, and every element of $W_{k,u}$ $(u < r)$ is $< N_{k,r}$ with residue $< D_{0,k}/2$ modulo $P_k$. Also $J_k, H_k \subseteq [0, N_{k,r})$, and $(H_k \cap A_t) \cap H_k(\gamma_r) = \emptyset$ because $H_k(\gamma_r) \subseteq A_o$ and $A_t \cap A_o = \emptyset$. Thus every $x \in A_t \cap [0, N)$ satisfies either $x \in W_{k,r}$, or $x < N_{k,r}$ with one of:

$$\text{(a) } x \bmod P_k < D_{0,k}/2, \qquad \text{(b) } x \in J_k, \qquad \text{(c) } x \in H_k \setminus H_k(\gamma_r).$$

Now let $x, y \in A_t$ with $z = x + y < N$:

(1) If $x, y < N_{k,r}$, then $z \in E_1 := [0, 2N_{k,r})$, which has size $2N_{k,r} = N/(2m_k)$.
(2) If both $x \bmod P_k < D_{0,k}/2$ and $y \bmod P_k < D_{0,k}/2$ (covering $x, y \in W_{k,r}$ and case (a) paired with $W_{k,r}$), then $z \bmod P_k < D_{0,k}$, placing $z$ in $E_2 := \{z < N : z \bmod P_k < D_{0,k}\}$ of size $(N/P_k)D_{0,k} = N/(2m_k)$.
(3) Otherwise one summand (say $y$) lies in $W_{k,r}$ and the other ($x$) satisfies (b) or (c). If $x \in J_k$, then $Q_k \mid L_{0,k} \mid x$, so $\Lambda_{\gamma_r}(x + y) = \Lambda_{\gamma_r}(y) \equiv 0 \pmod{m_k}$. If $x = h_k(j, u) \in H_k \setminus H_k(\gamma_r)$, then $(j, u)$ is outside the image of $\gamma_r$, so $c_{\gamma_r}(j, u) = 0$ and (1) gives $\Lambda_{\gamma_r}(x + y) = \Lambda_{\gamma_r}(y) + 0 \equiv 0 \pmod{m_k}$. Hence $z$ lies in $E_3 := \{z < N : m_k \mid \Lambda_{\gamma_r}(z)\}$. Since $B_k \mid N$ and $m_k \mid B_k$, each of the $N/B_k$ values of $\lfloor z/B_k \rfloor$ admits exactly $B_k/m_k$ residues $z \bmod B_k$ with $m_k \mid \Lambda_{\gamma_r}(z)$, so $|E_3| = N/m_k$.

Therefore $|(A_t + A_t) \cap [0, N)| \leq |E_1| + |E_2| + |E_3| = \frac{N}{2m_k} + \frac{N}{2m_k} + \frac{N}{m_k} = \frac{2N}{m_k}$. □

*Proof of the Theorem.* By Lemma 2, $\underline{d}(A + A) \geq 3/256 > 0$. Suppose $A = A_1 \sqcup A_2$ with $\underline{d}(A_1 + A_1) > 0$ and $\underline{d}(A_2 + A_2) > 0$. Choose integers $K \geq 1$ and $N_* \geq 1$ such that $|(A_i + A_i) \cap [0, n)| \geq n/K$ for both $i = 1, 2$ and all $n \geq N_*$. Pick $k = \max(N_*, K)$, so that $m_k = 2k+4 > 2K$ and $N_{0,k} > M_k \geq k \geq N_*$. By Lemma 3, some $r < R_k$ satisfies $H_k(\gamma_r) \subseteq A_o$ for one piece $A_o \in \{A_1, A_2\}$. Letting $A_t$ be the other piece and $N = N_{k,r+1} \geq N_{0,k} \geq N_*$, Lemma 4 gives

$$\frac{N}{K} \leq |(A_t + A_t) \cap [0, N)| \leq \frac{2N}{m_k} < \frac{N}{K},$$

a contradiction. □

*Remark* (How the parameters fit together).

- Choosing $B_k = 8m_k^3$ ensures both $m_k \mid B_k$ (so $\Lambda_\gamma(z) \bmod m_k$ is uniform over $z \bmod B_k$ for every fixed $\lfloor z/B_k \rfloor$) and $d_{0,k}/B_k = 2m_k^2/(8m_k^3) = 1/(4m_k) \leq 1/16$, so forbidding the top digit $B_k - 1$ across all $d_{0,k}$ coordinate digits keeps at least $(1 - 1/B_k)^{d_{0,k}} \geq 3/4$ of the cube $[0, B_k^{d_{0,k}})$.
- The divisibility $Q_k \mid L_{0,k} \mid (D_{0,k}/2) \mid D_{0,k} \mid P_k \mid N_{k,r}$ decouples the coarse scale arithmetic modulo $P_k$ and $D_{0,k}$ from the fine base-$B_k$ linear form $\Lambda_\gamma$ (which is $Q_k$-periodic).
- Setting $N_{k,r+1} = 4m_k N_{k,r}$ makes the prefix leakage $2N_{k,r} = N_{k,r+1}/(2m_k)$ negligible ($O(1/m_k)$) at the top of window $r$, while losing only a bounded factor of 4 in $\underline{d}(A + A)$ across the transition $[N_{k,r}, 4N_{k,r})$.

The argument above is formalized in Lean 4 in `FormalConjectures/ErdosProblems/741.lean` (theorem `Erdos741.erdos_741.variants.lower`).

## References


[1] P. Erdős, *Some problems in number theory, combinatorics and combinatorial geometry*, Math. Pannon. **5** (1994), 261–269.

[2] *Erdős Problem #741*, `https://www.erdosproblems.com/741`.

[3] The Formal Conjectures Authors, *FormalConjectures/ErdosProblems/741.lean*, `https://github.com/google-deepmind/formal-conjectures`.